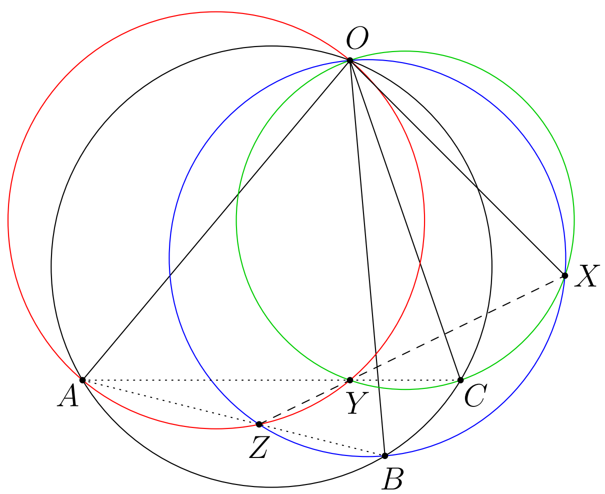
Collinearity - Solutions

Problem 1. (Mexico MO 1993 Problem 5) OA, OB, OC are three chords of a circle. The circles with diameters OA, OB meet again at Z, the circles with diameters OB, OC meet again at X, and the circles with diameters OC, OA meet again at Y. Show that X, Y, Z are collinear.

Solution.



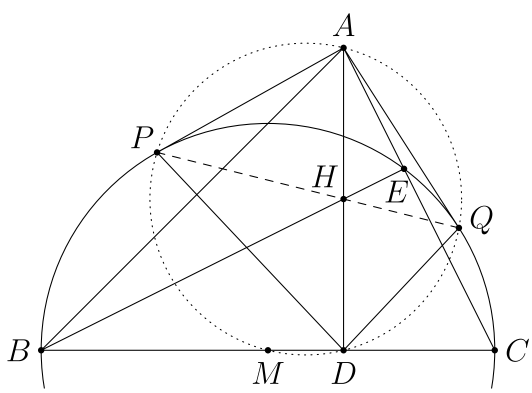
From the diameters, we have $\angle OYC = \angle OYA = 90°$. So A, Y, C are collinear. Similarly, A, Z, B are collinear (so are B, C, X). Therefore, we have

$$\angle OXY = \angle OCY = \angle OCA = \angle OBA = \angle OBZ = \angle OXZ.$$

This implies X, Y, Z are collinear. (This proof only works for this configuration. Alternatively, one may regard these angles as directed angles.)

Problem 2. (CMO 1996 Problem 1) Let H be the orthocentre of an acute $\triangle ABC$. The tangents AP, AQ are constructed from A to the circle with diameter BC. The contact points are P and Q respectively. Prove that P, H, Q are collinear.

Solution.

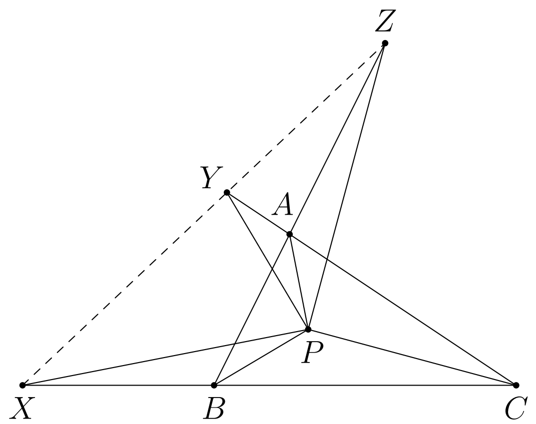


Let D and E be the feet of the altitudes from A and B respectively, and let M be the midpoint of BC. Note that E lies on the circle with diameter BC, and that H, D, C, E are concyclic. So we have $AQ^2 = AE \times AC = AH \times AD$, which implies $\angle AQH = \angle ADQ$.

Next, since we have $\angle APM = \angle ADM = \angle AQM = 90°$, the points A, P, M, D, Q are concyclic. Together with $AP = AQ$, we know that AD is the internal angle bisector of $\angle PDQ$. Therefore, we have $\angle AQP = \angle ADP = \angle QDA$. It follows that $\angle AQH = \angle AQP$, and hence P, H, Q are collinear.

Problem 3. Let P be a point inside $\triangle ABC$. The line through P perpendicular to AP meets the sideline BC at X. The line through P perpendicular to BP meets the sideline CA at Y. The line through P perpendicular to CP meets the sideline AB at Z. Prove that X, Y, Z are collinear.

Solution.



We only prove the configuration as shown as the general case is similar. Using ratio of areas, we have

$$\frac{BX}{XC} = \frac{[PBX]}{[PCX]} = \frac{\frac{1}{2}PB \times PX \sin\angle BPX}{\frac{1}{2}PC \times PX \sin\angle CPX} = \frac{PB}{PC} \times \frac{\sin(\angle APB - 90°)}{\sin(270° - \angle APC)} = \frac{PB}{PC} \times \frac{\cos\angle APB}{\cos\angle APC}.$$

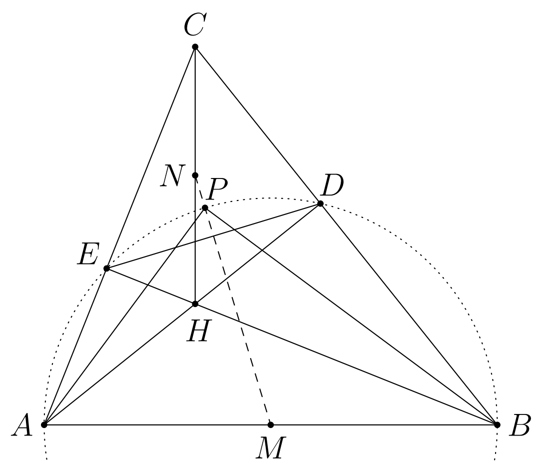
Similarly, we have $\dfrac{CY}{YA} = \dfrac{PC}{PA} \times \dfrac{\cos\angle BPC}{\cos\angle BPA}$ and $\dfrac{AZ}{ZB} = \dfrac{PA}{PB} \times \dfrac{\cos\angle CPA}{\cos\angle CPB}$. So

$$\frac{BX}{XC} \times \frac{CY}{YA} \times \frac{AZ}{ZB} = 1.$$

From the configuration and Menelaus' theorem, X, Y, Z are collinear.

Problem 4. (Bosnia and Herzegovina TST 2005 Problem 1) Let H be the orthocentre of an acute triangle ABC. Prove that the midpoints of AB and CH and the intersection point of the angle bisectors of $\angle CAH$ and $\angle CBH$ lie on the same line.

Solution.

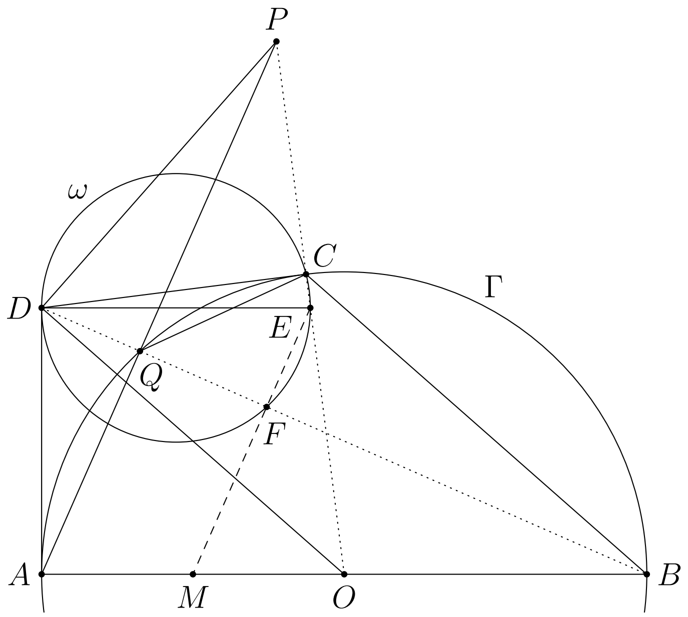


Let M and N be the midpoints of AB and CH respectively. Let P be the intersection of the internal angle bisectors of $\angle CAH$ and $\angle CBH$. Also, let AD and BE be the altitudes. Firstly, C, E, H, D lie on the circle with diameter CH, where N is the centre of the circle. So $ND = NE$. Secondly, A, B, D, E lie on the circle with diameter AB, where M is the centre of the circle. So $MD = ME$.

Next, since $\angle DAE = \angle DBE$, we have $\angle DBP = \angle EBP = \angle DAP = \angle EAP$. These show P lies on the circle through A, B, D, E, and P is the midpoint of the arc DE. So $PD = PE$. Therefore, N, M, P lie on the perpendicular bisector of DE, and hence they are collinear.

Problem 5. Let AB be a diameter of a circle Γ and let O be the centre of Γ. Let C be an arbitrary point on Γ. The tangents at A and C to Γ intersect at a point D. A circle ω passing through C is tangent to the line AD at D. Let DE be a diameter of ω. The line BD meets ω again at F. Let M be the midpoint of OA. Prove that E, F, M are collinear.

Solution.



Let P be the symmetric point of O with respect to E. Since $AOCD$ is a kite and $DE//AO$ (as both DE and AO are perpendicular to AD), we have $\angle EDO = \angle DOA = \angle EOD$. This shows $ED = EO = EP$, and hence $\angle PDO = 90^\circ$.

Let Q be the intersection of AP and Γ. Then we have

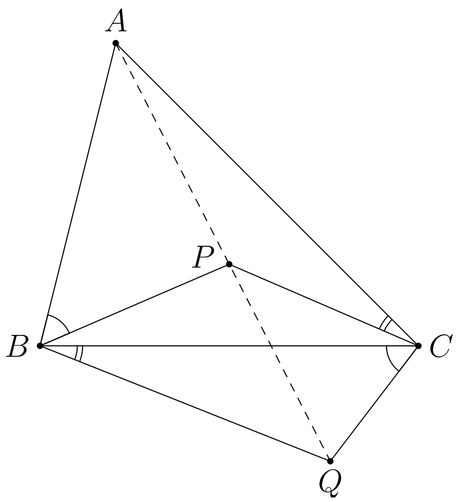
$$\angle CQP = \angle CBA = \frac{1}{2}\angle COA = \angle COD = 90^\circ - \angle CDO = \angle CDP.$$

This implies P, D, Q, C are concyclic. It follows that $\angle PQD = \angle PCD = 90^\circ$. As we also have $\angle AQB = 90^\circ$, the points D, Q, B are collinear. Thus, BD and AP are perpendicular.

By the midpoint theorem, we have $PA//EM$. So $EM \perp BD$. Note that $\angle EFD = 90^\circ$. This implies E, F, M are collinear.

Problem 6. Given acute $\triangle ABC$, let P be a point on the perpendicular bisector of the side BC such that P and A lie on the same side of the line BC. Let Q be a point on the opposite side of the line BC as P such that $\angle BCQ = \angle ABP$ and $\angle CBQ = \angle ACP$. Prove that A, P, Q are collinear.

Solution.



By the trigonometric form of Ceva's theorem, we have

$$\frac{\sin\angle BAP}{\sin\angle CAP} \times \frac{\sin\angle ACP}{\sin\angle BCP} \times \frac{\sin\angle CBP}{\sin\angle ABP} = 1.$$

Since $\angle BCP = \angle CBP$, this implies

$$\frac{\sin\angle BAP}{\sin\angle CAP} = \frac{\sin\angle BCQ}{\sin\angle CBQ} = \frac{BQ}{CQ}.$$

Consider $\triangle ABQ$ and $\triangle ACQ$. We have

$$\frac{BQ}{\sin\angle BAQ} = \frac{AQ}{\sin\angle ABQ} \quad \text{and} \quad \frac{CQ}{\sin\angle CAQ} = \frac{AQ}{\sin\angle ACQ}.$$

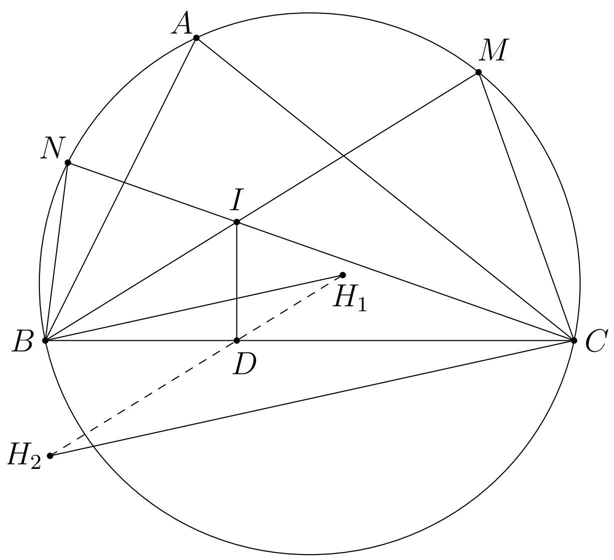
Note that $\angle ABQ = \angle ACQ$. So these ratios are equal and hence

$$\frac{\sin \angle BAP}{\sin \angle CAP} = \frac{BQ}{CQ} = \frac{\sin \angle BAQ}{\sin \angle CAQ}.$$

As $\triangle ABC$ is acute, this implies A, P, Q are collinear.

Problem 7. Let I be the incentre of $\triangle ABC$. Let D be the contact point of the incircle of $\triangle ABC$ and the side BC. Denote by H_1 and H_2 the orthocentres of $\triangle AIB$ and $\triangle AIC$ respectively. Prove that H_1, H_2, D are collinear.

Solution.



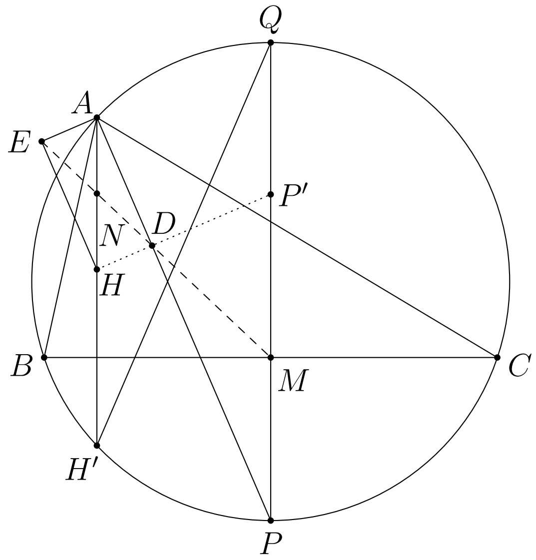
Since both BH_1 and CH_2 are perpendicular to AI, they are parallel. Consider BH_1. It is the length from a vertex of $\triangle ABI$ to its orthocentre. So we have $BH_1 = 2R_1 \cos \angle ABI$, where R_1 is the circumradius of $\triangle ABI$. Let BI and CI meet the circumcircle of $\triangle ABC$ at M and N respectively. Recall that N is the circumcentre of $\triangle ABI$. Thus, we have $R_1 = BN$. It follows that

$$\frac{BD}{BH_1} = \frac{BD}{2BN \cos \frac{B}{2}} = \frac{BD}{2BN \cos \angle DBI} = \frac{BI}{2BN}.$$

Similarly, we have $\dfrac{CD}{CH_2} = \dfrac{CI}{2CM}$. This is equal to $\dfrac{BI}{2BN}$ since $\triangle NBI \sim \triangle MCI$. Therefore, we find that $\triangle DBH_1 \sim \triangle DCH_2$. Thus, H_1, D, H_2 are collinear.

Problem 8. (Bosnia and Herzegovina MO 2018 Grade 9 Problem 5) Let H be the orthocentre of an acute triangle ABC and M the midpoint of side BC. If D and E are the feet of perpendicular of H on the internal and external angle bisectors of $\angle BAC$, prove that M, D and E are collinear.

Solution.

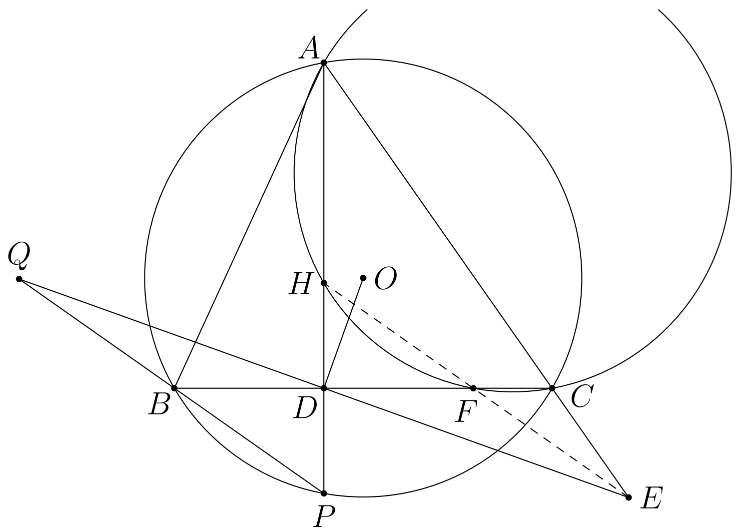


Let AD meet the circumcircle of $\triangle ABC$ at P. Let H' and P' be the reflection of H and P in the line BC respectively, and let PQ be a diameter of the circumcircle of $\triangle ABC$. Note that both $AH'PQ$ and $HH'PP'$ are isosceles trapezoid, and BC is perpendicular to their parallel sides. So HP' and AQ are parallel, and hence $HP' \perp AP$. This implies H, D, P' are collinear.

Now, as $AH//PP'$, the midpoint N of AH, the midpoint M of PP', and the point D are collinear. Since $AEHD$ is a rectangle, N is also the midpoint of DE. This shows E, D, M are collinear.

Problem 9. Let O and H be the circumcentre and orthocentre of $\triangle ABC$ respectively. Let D be the foot of the altitude from A. The line through D perpendicular to OD meets the extension of AC at E. The circumcircle of $\triangle CAH$ intersects the line BC again at F. Prove that H, E, F are collinear.

Solution.



Let AD meet the circumcircle of $\triangle ABC$ again at P. Note that D is the midpoint of HP. Let BP meet DE at Q. By the butterfly theorem (applying to the chords AP and BC), we have $DQ = DE$. Thus, HP and QE bisect each other. This shows $QPEH$ is a parallelogram. In particular, we have $EH//PQ$.

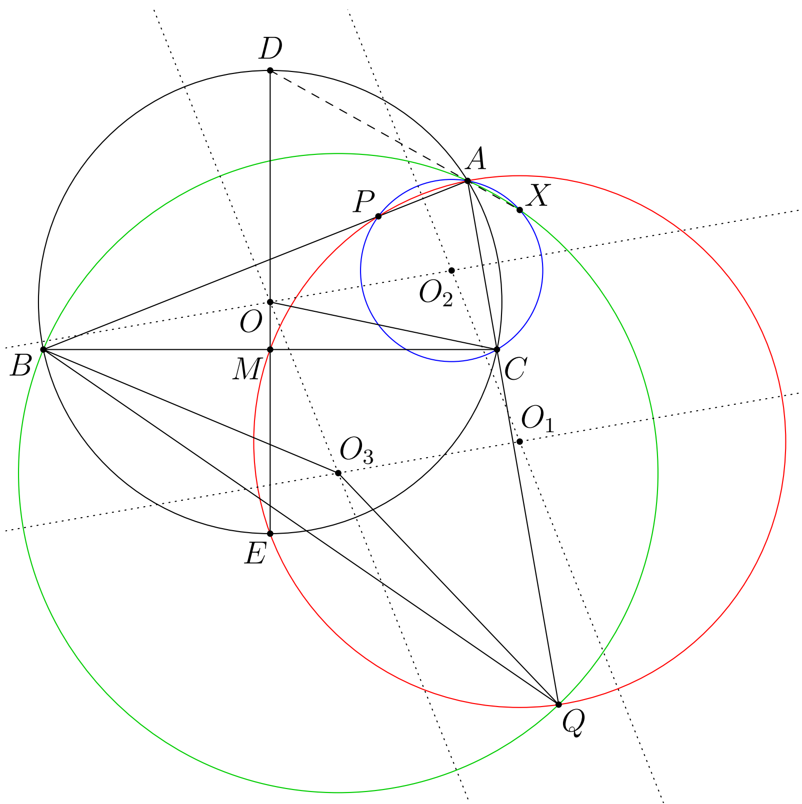
In addition, we have

$$\angle FHP = \angle ACB = \angle APB.$$

This shows $FH//PB$, and hence H, F, E are collinear.

Problem 10. In $\triangle ABC$, $AB > AC$. Let M be the midpoint of BC. The perpendicular bisector of BC intersects the circumcircle of $\triangle ABC$ at the points D and E, where D lies on the same side of BC as A. The circumcircle of $\triangle AME$ meets the line AB and AC again at P and Q respectively. The circumcircles of $\triangle ACP$ and $\triangle ABQ$ meet again at X. Prove that D, A, X are collinear.

Solution.



Let O, O_1, O_2, O_3 be the circumcentres of $\triangle ABC$, $\triangle AME$, $\triangle APC$ and $\triangle ABQ$ respectively. Since O and O_3 lie on the perpendicular bisector of AB, O_1 and O_2 lie on the perpendicular bisector of AP, we have $OO_3//O_2O_1$. Similarly, we have $OO_2//O_3O_1$. So $OO_3O_1O_2$ is a parallelogram.

As $\angle BOC = 2\angle BAC = 2\angle BAQ = \angle BO_3Q$, the isosceles $\triangle BOC$ and $\triangle BO_3Q$ are similar. Thus, we have $\triangle BOO_3 \sim \triangle BCQ$ (this is indeed a well-known fact from spiral similarity). This implies $\dfrac{OO_3}{CQ} = \dfrac{BO}{BC}$. Similarly, we have $\triangle COO_2 \sim \triangle CBP$, and hence $\dfrac{OO_2}{BP} = \dfrac{CO}{CB}$. The above two ratios are equal since $BO = CO$.

Next, from $\angle EBP = \angle ECQ$, $\angle EPB = \angle EQC$ and $EB = EC$, we have $\triangle EBP \cong \triangle ECQ$. This yields $BP = CQ$. Together with $\dfrac{OO_3}{CQ} = \dfrac{OO_2}{BP}$, we get $OO_3 = OO_2$. This shows $OO_3O_1O_2$ is a rhombus. So the diagonals OO_1 and O_3O_2 are perpendicular. Thus, the radical axis AE of the circumcircles of $\triangle ABC$ and $\triangle AME$ is perpendicular to the radical axis AX of the circumcircles of $\triangle APC$ and $\triangle ABQ$. This means $\angle EAX = 90°$. As $\angle EAD = 90°$, D, A, X are collinear.